

IT TROUBLESHOOTING CASE STUDY

Audio Volume Attenuation — HP ENVY x360

Technician: Mahi | Date: June 10, 2026 | Ticket #: MAHI-2026-001

1. Ticket Summary

Device	HP ENVY x360 Laptop
OS	Windows 11
Audio Hardware	Realtek(R) Audio + Intel Smart Sound Technology (SST)
OEM Software	Bang & Olufsen Audio Control
Output Device	Built-in laptop speakers
Issue Duration	Several months (persistent)
Priority	Medium — Functional but degraded audio output

2. Problem Description

The user reported that the laptop speaker volume was abnormally low despite the system volume being set to maximum (100%). All audio sources were affected — YouTube, streaming, system sounds, and all applications produced uniformly low volume.

Key Symptoms Reported:

- Volume extremely low at maximum system setting across all applications
- When muting and unmuting, the original full volume was audible for a fraction of a second before dropping back to the attenuated level
- Volume would occasionally and inconsistently return to normal levels for approximately one minute during media playback, then drop again
- The issue had persisted for several months with no clear trigger event

Technical Name: Automatic Volume Attenuation (Audio Ducking)

The mute/unmute behavior was a critical diagnostic indicator — the brief burst of full volume confirmed the speakers and audio hardware were physically capable of full output, meaning the attenuation was occurring in a software processing layer that needed a fraction of a second to re-engage after unmute.

3. Diagnostic Process

The troubleshooting followed a systematic elimination approach, testing each layer of the audio stack from the outermost software processing down to the hardware. Each step was tested individually with audio playback before proceeding.

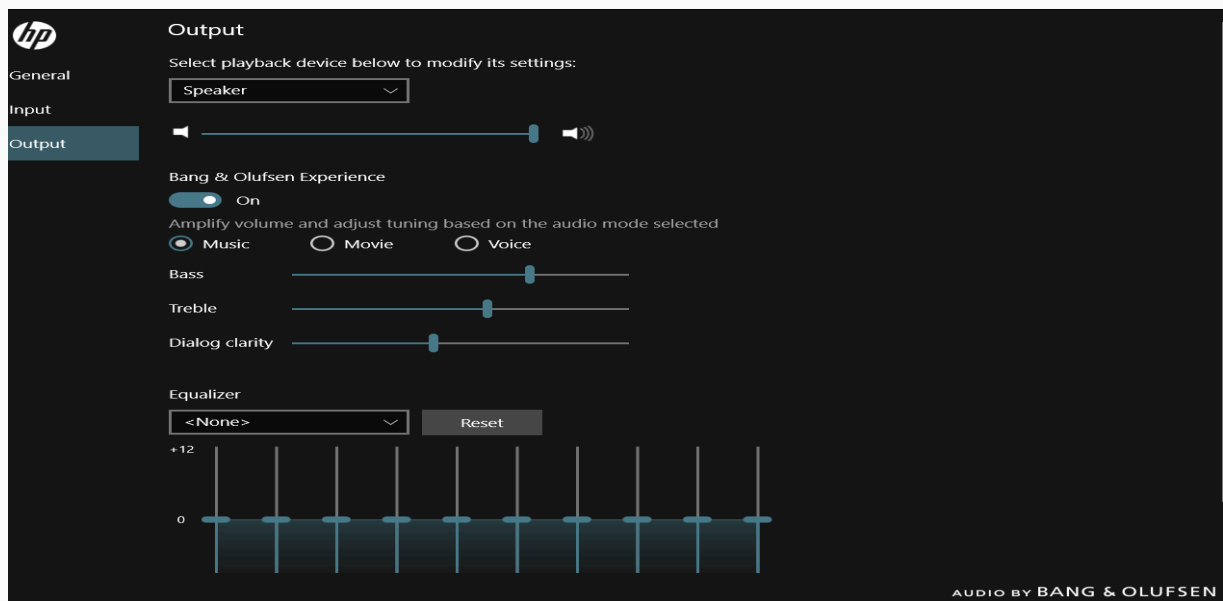
Step 1: Windows Communications Ducking

Checked the Windows Sound Control Panel > Communications tab. This feature automatically reduces system volume by up to 80% when Windows detects voice/communications activity (Teams, Zoom, Discord). The setting was found at "Reduce the volume of other sounds by 80%" and was changed to "Do nothing."

RESULT: No change — Volume remained low. Communications ducking ruled out.

Step 2: Bang & Olufsen Audio Control — B&O Experience

Opened HP's pre-installed Bang & Olufsen Audio Control application. The Output tab revealed the "Bang & Olufsen Experience" toggle was set to ON with the description "Amplify volume and adjust tuning based on the audio mode selected." This processing layer intercepts the audio stream and applies dynamic volume adjustment based on the selected mode (Music/Movie/Voice). The toggle was switched to OFF.

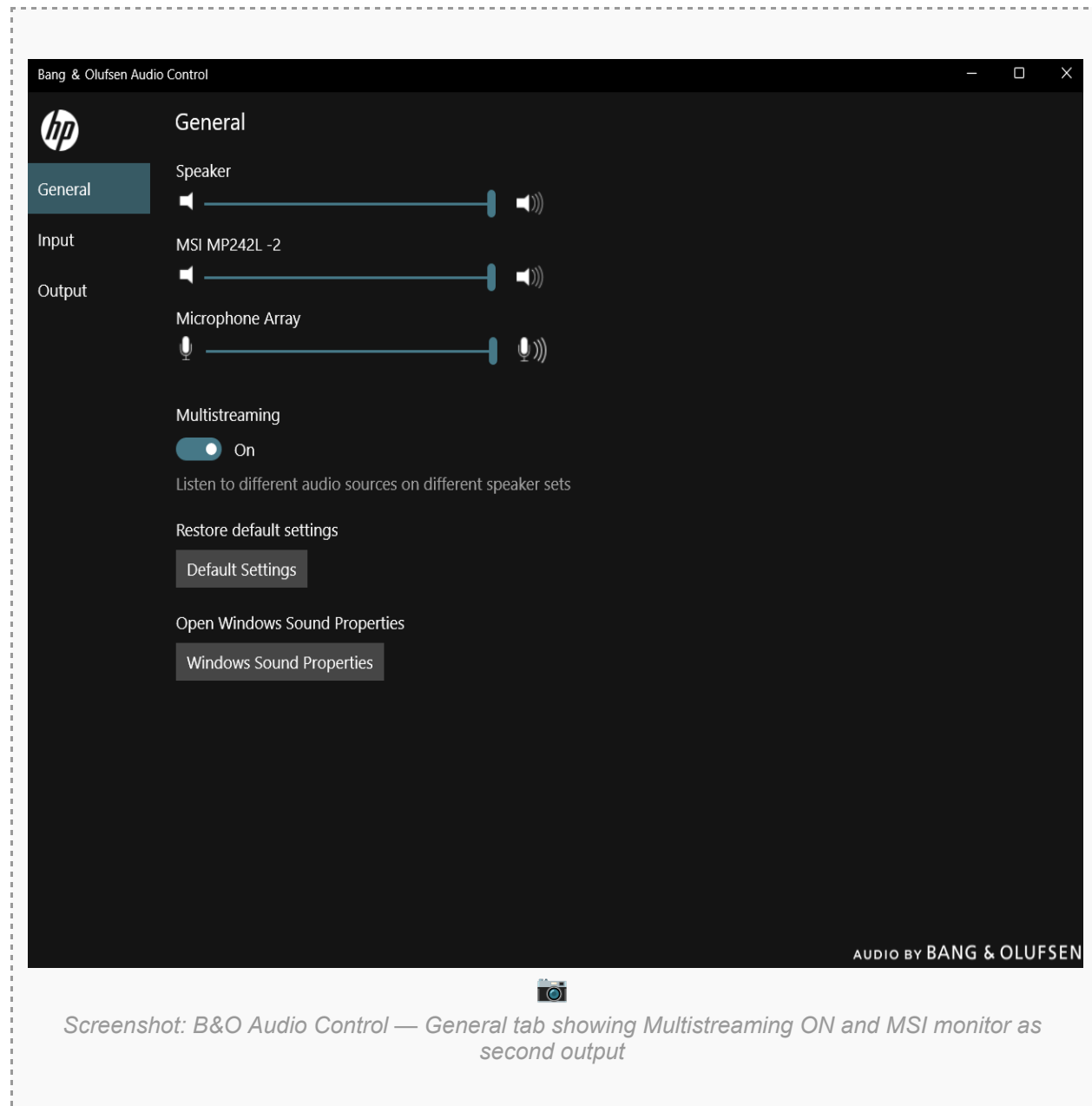


Screenshot: B&O Audio Control — Output tab showing "Bang & Olufsen Experience" toggle ON

RESULT: No change — Volume remained low. B&O Experience processing ruled out.

Step 3: B&O Multistreaming

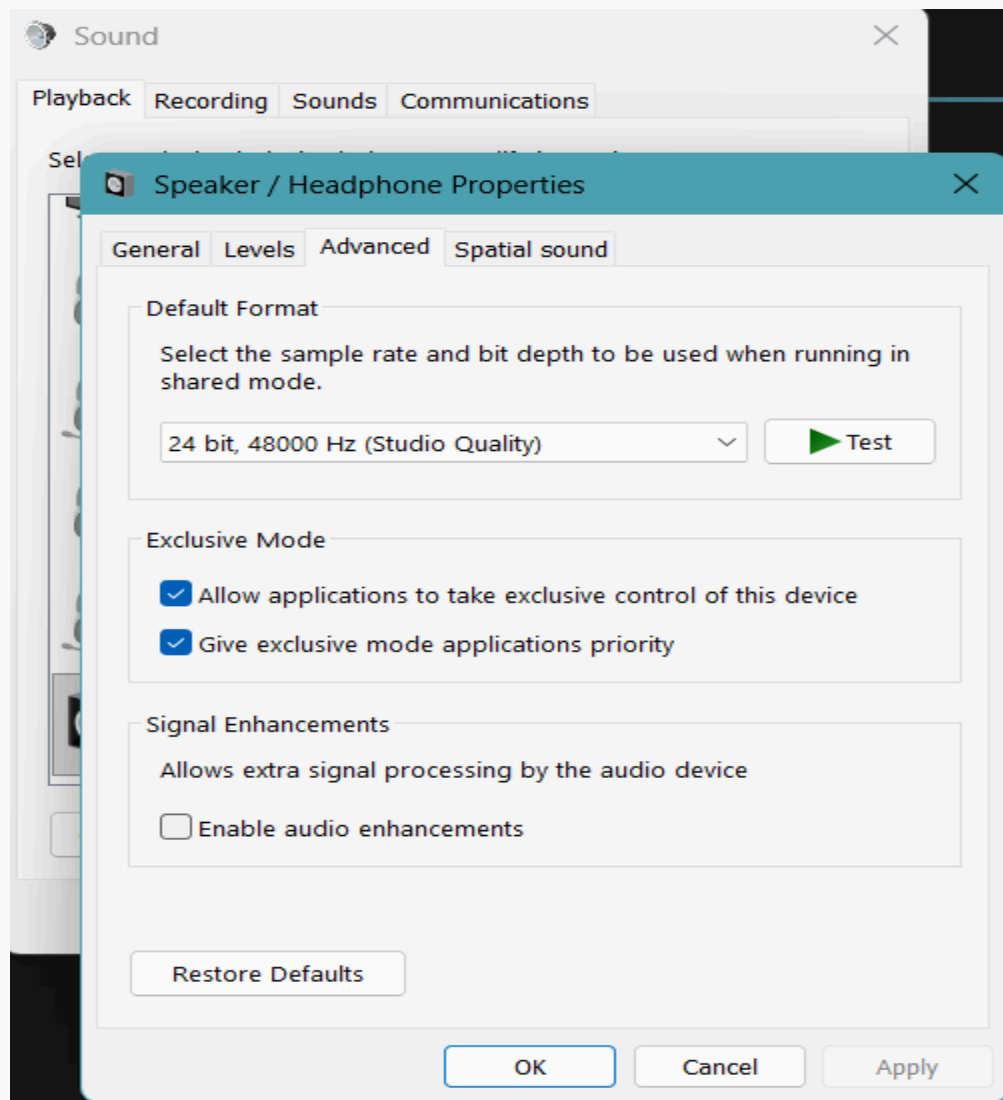
On the B&O General tab, the "Multistreaming" feature was found ON. This feature splits the audio signal across multiple output devices simultaneously (laptop speakers + MSI MP242L-2 external monitor). When active, the signal is divided between outputs, which can reduce volume to each individual device. The toggle was switched to OFF.



RESULT: No change — Volume remained low. Multistreaming signal splitting ruled out.

Step 4: Windows Audio Enhancements

Opened Windows Sound Properties > Playback > Speaker/Headphone Properties > Advanced tab. Checked the "Signal Enhancements" section — the "Enable audio enhancements" checkbox was already unchecked (disabled). Also verified the Default Format was set to "24 bit, 48000 Hz (Studio Quality)" which is correct.

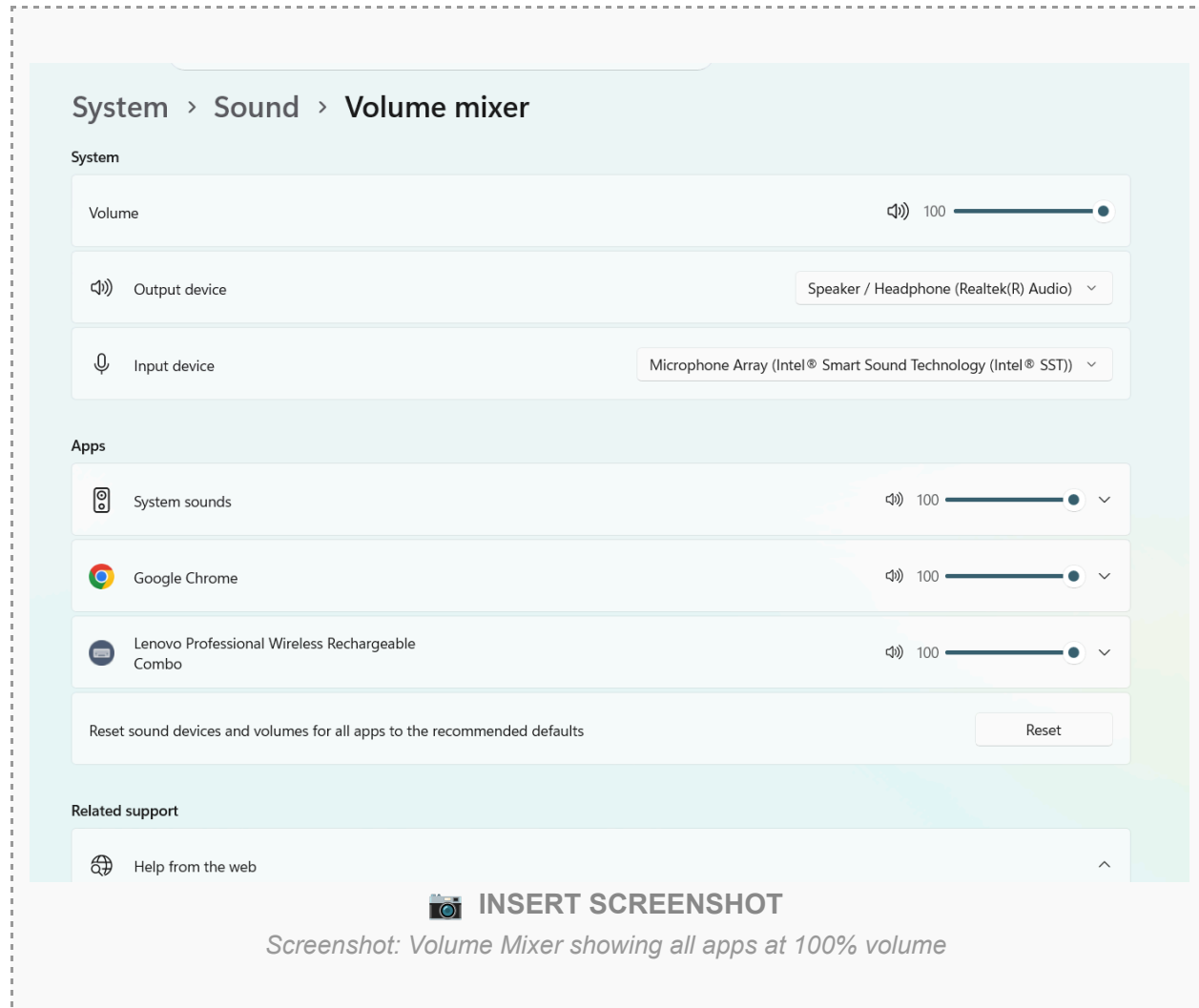


Screenshot: Speaker/Headphone Properties — Advanced tab showing enhancements disabled

RESULT: Already disabled — Windows audio enhancements were not the cause.

Step 5: Volume Mixer App-Level Check

Opened Windows Settings > System > Sound > Volume Mixer. Verified all volume levels: System Volume at 100, System Sounds at 100, Google Chrome at 100, and all other listed applications at 100. Output device confirmed as "Speaker / Headphone (Realtek(R) Audio)." No individual app was being throttled.



RESULT: All levels at maximum — App-level volume throttling ruled out.

Step 6: Headphone Isolation Test — PIVOTAL DIAGNOSTIC MOMENT

Connected earbuds/headphones to the laptop and tested audio playback. The audio through headphones was at full normal volume. This was the critical test that changed the entire diagnostic direction.

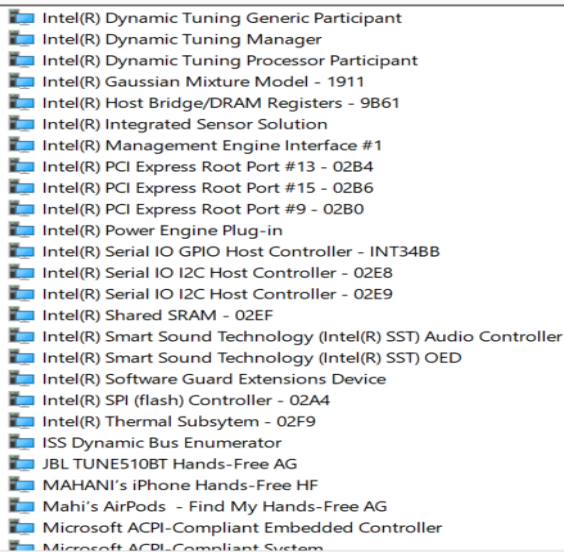
Why this test matters:

Both the internal laptop speakers and the headphone jack share the exact same software audio pipeline — same Realtek driver, same Windows audio engine, same Volume Mixer, same enhancement settings. If headphones produce correct volume but internal speakers do not, the problem cannot be in the shared software stack. The issue must exist in the output path that is unique to the internal speakers — either the speaker-specific driver gain profile or the physical speaker amplifier hardware.

RESULT: Headphones normal, speakers low — Problem isolated to internal speaker output path. Software audio processing stack fully eliminated.

Step 7: Intel Smart Sound Technology (SST) Disable Test

In Device Manager > System Devices, identified two Intel SST components: "Intel(R) Smart Sound Technology (Intel(R) SST) Audio Controller" and "Intel(R) Smart Sound Technology (Intel(R) SST) OED." The Audio Controller (the one in the audio output path) was disabled to test whether SST's DSP processing was causing the speaker-specific attenuation.



Intel(R) Dynamic Tuning Generic Participant
Intel(R) Dynamic Tuning Manager
Intel(R) Dynamic Tuning Processor Participant
Intel(R) Gaussian Mixture Model - 1911
Intel(R) Host Bridge/DRAM Registers - 9B61
Intel(R) Integrated Sensor Solution
Intel(R) Management Engine Interface #1
Intel(R) PCI Express Root Port #13 - 02B4
Intel(R) PCI Express Root Port #15 - 02B6
Intel(R) PCI Express Root Port #9 - 02B0
Intel(R) Power Engine Plug-in
Intel(R) Serial IO GPIO Host Controller - INT34BB
Intel(R) Serial IO I2C Host Controller - 02E8
Intel(R) Serial IO I2C Host Controller - 02E9
Intel(R) Shared SRAM - 02EF
Intel(R) Smart Sound Technology (Intel(R) SST) Audio Controller
Intel(R) Smart Sound Technology (Intel(R) SST) OED
Intel(R) Software Guard Extensions Device
Intel(R) SPI (flash) Controller - 02A4
Intel(R) Thermal Subsystem - 02F9
ISS Dynamic Bus Enumerator
JBL TUNE510BT Hands-Free AG
MAHANI's iPhone Hands-Free HF
Mahi's AirPods - Find My Hands-Free AG
Microsoft ACPI-Compliant Embedded Controller
Microsoft ACPI-Compliant System

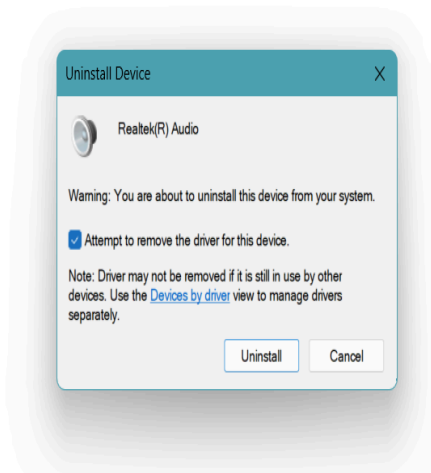
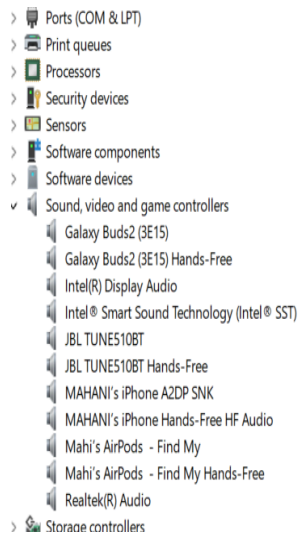
Screenshot: Device Manager — System Devices showing both Intel SST entries

RESULT: All audio output was lost — Intel SST is a required audio component on this HP model, not an optional processor. SST was re-enabled. This confirmed SST is integral to the audio pipeline but could not be isolated as the cause.

Step 8: Clean Realtek Audio Driver Reinstall — RESOLUTION

In Device Manager > Sound, video and game controllers, right-clicked "Realtek(R) Audio" and selected "Uninstall device" with the "Attempt to remove the driver for this device" checkbox enabled. This fully removes the existing driver and its stored configuration (including the corrupted speaker gain profile). The laptop was then restarted.

Upon reboot, Windows automatically detected the audio hardware and installed a fresh driver. The internal speaker volume immediately returned to full normal levels.



✓ **RESOLVED — Speaker volume restored to full normal output. Clean driver reinstall eliminated the corrupted speaker gain profile.**

4. Root Cause Analysis

The Realtek audio driver maintained separate gain configuration profiles for internal speakers versus the headphone output jack. The speaker-specific gain profile had become corrupted at some point in the preceding months, causing the driver to apply a significantly reduced gain level to the internal speaker amplifier while leaving the headphone output unaffected.

The intermittent behavior (brief bursts of full volume, occasional minutes of correct audio) was consistent with a corrupted driver profile that periodically failed to load its attenuated setting, momentarily allowing the raw full-volume signal through before the corrupted profile re-engaged.

The clean uninstall (with driver removal) deleted the corrupted configuration data. The fresh driver installed on reboot created new, default gain profiles for both output paths, restoring correct speaker volume.

5. Resolution Summary

Root Cause	Corrupted Realtek driver speaker gain profile
Fix Applied	Clean uninstall of Realtek Audio driver with driver removal, followed by system restart for automatic fresh driver installation
Verification	Internal speaker volume confirmed at full normal output across all applications
Status	RESOLVED

6. Post-Fix Monitoring

If the issue recurs, the following should be investigated as potential causes of re-corruption:

- Windows Update: Check Settings > Windows Update > Update History for any recently pushed Realtek driver updates that may reintroduce the corrupted profile
- Bang & Olufsen Software: Verify the B&O Experience toggle has not been re-enabled, as it may attempt to re-apply its processing layer over the fresh driver
- Intel SST Interaction: The Intel Smart Sound Technology DSP may re-negotiate its processing parameters with the new Realtek driver over time

If recurrence is confirmed, the recommended permanent fix is to block automatic Realtek driver updates through Windows Device Installation Settings or Group Policy to prevent the corrupted driver from being redownloaded.

7. Diagnostic Methodology Notes

This case demonstrated a systematic layer-by-layer elimination approach to audio troubleshooting:

1. Started at the outermost software layer (Windows Communications setting) and worked inward
2. Tested OEM processing software (B&O Audio Control) as a separate layer
3. Verified OS-level controls (Volume Mixer, Enhancements, Spatial Sound)
4. Used the headphone isolation test to definitively separate shared software pipeline from speaker-specific path — this was the pivotal diagnostic step
5. Tested the DSP hardware layer (Intel SST) to determine if it was optional or required
6. Performed clean driver reinstall as the final software-level intervention before escalating to hardware diagnosis

Key Takeaway:

The headphone test is the single most valuable diagnostic step in laptop audio troubleshooting. It takes 10 seconds and immediately separates software-stack issues (affects all outputs) from speaker-path issues (affects only internal speakers). This test should be performed early in the diagnostic process to avoid unnecessary steps.